

# Smart Home Energy Consumption Dataset with Environmental & Occupancy Data

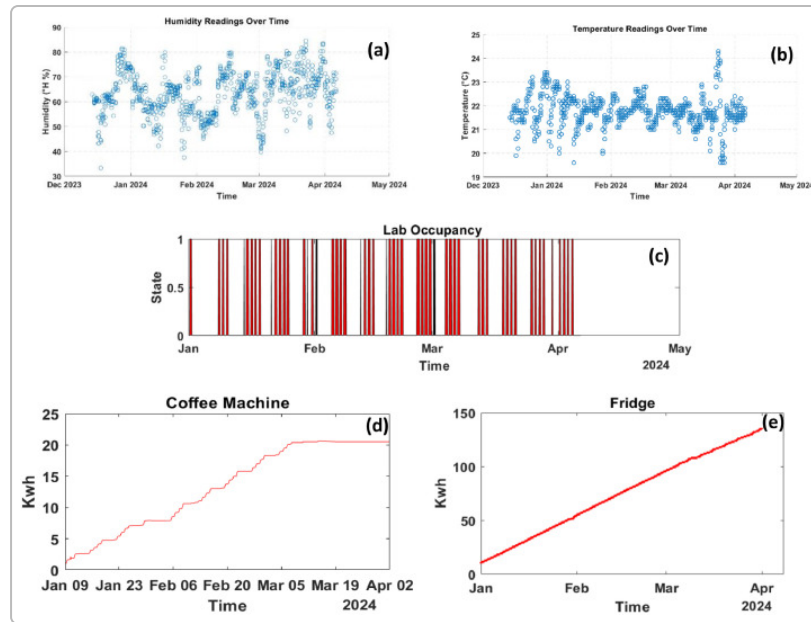
## Dataset Overview

One highly suitable open-access dataset comes from a smart building (analogous to a smart home) in the University of Sharjah's campus. This dataset provides detailed **appliance-level energy usage** recorded over time, alongside **environmental conditions** and **occupancy information**. It was collected as part of a smart campus initiative in early 2024 <sup>1</sup>. Key features of the dataset include:

- **Scope and Duration:** Data collected from **January 1 to June 20, 2024** in a campus building (offices, lab, and common areas), with **time-stamped readings** over roughly 6 months <sup>1</sup>. This covers daily operational patterns in a real-world smart home/building scenario.
- **Energy Consumption Data: Appliance-specific power usage** measurements (in watts, voltage, and cumulative kWh) for various devices (e.g. coffee machines, microwaves, refrigerators, lab equipment) <sup>2</sup> <sup>3</sup>. The dataset captures fine-grained consumption per device, enabling analysis of individual appliance behavior.
- **Environmental Parameters: Environmental sensor data** including temperature (°C) and humidity (%) readings in key rooms <sup>2</sup>. These factors are logged alongside energy usage to study how climate conditions correlate with energy consumption (e.g. HVAC or appliance performance relative to room temperature).
- **Occupancy Data: Room occupancy status** monitored via IoT motion sensors, providing timestamps of when rooms (labs, kitchen, etc.) are occupied or vacant <sup>2</sup> <sup>4</sup>. The dataset includes occupancy rates/presence indicators, a critical behavioral factor for energy use (since appliance use often depends on whether occupants are present).
- **Format and Accessibility:** Data is provided in easy-to-use CSV files with timestamps and respective sensor readings. The dataset is published for academic research under an **open CC BY 4.0 license** <sup>5</sup>, meaning it can be freely used with attribution. It is **publicly downloadable** and documented in a peer-reviewed Data in Brief article <sup>1</sup>.

### Access and Resources:

- **GitHub Repository (Download Data):** The full dataset (raw CSV files for power, occupancy, and environment readings) is available on GitHub – *Dataset of IoT-Based Energy and Environmental Parameters in a Smart Building Infrastructure* by University of Sharjah researchers. (Direct link to repository: <https://github.com/adel8641/Dataset-of-IoT-Based-Energy-and-Environmental-Parameters-in-a-Smart-Building-Infrastructure>).
- **Dataset Reference (Publication):** *Data in Brief*, vol. 56, 2024, Article 110769 – Oulefki et al., “**Dataset of IoT-based energy and environmental parameters in a smart building infrastructure.**” This open-access paper describes the data collection and contents in detail <sup>1</sup> (DOI: 10.1016/j.dib.2024.110769).



**Figure 1:** Sample data from the smart building dataset, showing multiple sensor readings over time <sup>6</sup> <sup>7</sup> . Subplots (a) and (b) plot humidity (%) and temperature (°C) trends in a laboratory room; (c) indicates the lab's occupancy state (1 = occupied, 0 = empty) over the same timeline; (d) and (e) depict the cumulative energy usage (in kWh) of a **coffee machine** and a **fridge**, respectively. This example highlights how environmental conditions and occupancy correlate with appliance energy consumption. Such rich, multi-faceted data enables analysis of usage patterns and supports **machine learning** applications for energy efficiency.

## Potential Machine Learning Project Components

Using this dataset, you can develop **four complementary project components**, each incorporating a novel ML element. These components can be implemented in Python (for data analysis and modeling) and integrated with front-end or backend frameworks (e.g. a React dashboard or Node.js service) for a full-stack smart home solution:

1. **Energy Demand Forecasting:** Build a time-series ML model to **predict future energy consumption** for the home/building. Using historical appliance usage along with contextual features like time of day and environmental factors (temperature, humidity, etc.) can improve forecasting accuracy <sup>8</sup> . For example, a Python-based RNN/LSTM model or Facebook Prophet could predict the next-hour or next-day power usage for each appliance or the whole house. This prediction component could run on a backend server (Python/Node) and feed into a React front-end that visualizes upcoming energy demand, helping occupants plan and reducing peak usage. (*ML innovation: time-series prediction/regression with multi-variable sensor inputs.*)
2. **Occupancy Prediction & Behavior Modeling:** Leverage the occupancy sensor data together with appliance usage patterns to **infer or predict room occupancy and user behavior**. Since appliance on/off patterns are tightly linked with whether people are present <sup>9</sup> , a classification model (e.g. a decision tree or neural network) could be trained to detect occupancy from energy data in real-time. Conversely, you could predict when specific rooms are likely to be occupied based on past usage

routines (e.g. the lab is usually occupied in the afternoon alongside coffee machine use). Unsupervised learning (clustering) can also identify typical daily activity patterns. This component enables smart-home automation — for instance, automatically adjusting HVAC or lighting via a Python/Node backend when a room is predicted to be vacant, and providing behavior insights on a React UI. *(ML innovation: classification/clustering of usage data to model human occupancy patterns.)*

**3. Anomaly Detection in Appliance Usage:** Implement an ML-based monitoring system to **detect anomalies or inefficiencies** in energy consumption. Using unsupervised learning (such as Isolation Forest, one-class SVM, or autoencoder neural networks), the system can learn each appliance's normal power profile and flag deviations. For example, if the refrigerator's power draw spikes abnormally or a device is running at odd hours, the model would detect it as an outlier. The dataset's visualizations already highlight peak usage times and potential inefficiencies <sup>10</sup>, which you can quantify with ML. This anomaly detector could run as a Python service (or in Node.js via a Python ML microservice), issuing alerts to a front-end dashboard when something is amiss – e.g., notifying if an appliance might be malfunctioning or if lights were left on unexpectedly. *(ML innovation: unsupervised anomaly/outlier detection on time-series sensor data.)*

**4. Appliance-Level Load Disaggregation (NILM):** Use the rich appliance-level data to develop a **non-intrusive load monitoring** model. In practice, one often only has a total power meter for the whole home; here you can sum the individual device streams to simulate an aggregate signal, then train a model to **disaggregate** it and identify which appliances are running at what times. Techniques might include pattern recognition or deep learning (e.g. sequence-to-sequence models) that learn the “signature” of each device's power usage. By training on the provided labeled appliance data, the model could take an aggregate consumption curve and classify the contribution of each appliance. This component showcases an advanced ML application where a Python model (say, a deep neural network) could be deployed to a backend, and a React front-end could display real-time estimates of each appliance's consumption without needing separate hardware per appliance. *(ML innovation: multi-label classification or regression for signal separation, applying deep learning to energy data.)*

Each of these project components addresses a different aspect of smart home energy management using machine learning. By combining them, a final-year project could deliver a full intelligent energy system – with predictive analytics, smart sensing of occupancy, automated anomaly alerts, and insightful energy breakdowns – all powered by the **open dataset** above and integrated through Python-based ML and modern web technologies.

**Sources:** The dataset and its description are provided by Oulefki *et al.* (2024) <sup>1</sup>, including appliance-level energy, environmental, and occupancy data. Additional references highlight the role of environmental features in energy prediction <sup>8</sup> and the link between occupancy behavior and energy use <sup>9</sup>. The figure is reproduced from the open-access dataset publication <sup>6</sup> <sup>7</sup> to illustrate the data. All data is openly licensed (CC BY 4.0) for research use <sup>5</sup>.

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<sup>1</sup> <sup>2</sup> Dataset of IoT-based energy and environmental parameters in a smart building infrastructure - PubMed

<https://pubmed.ncbi.nlm.nih.gov/39676852/>

3 4 GitHub - adel8641/Dataset-of-IoT-Based-Energy-and-Environmental-Parameters-in-a-Smart-Building-Infrastructure: Data were collected using IoT sensors (energy consumption meters, temperature, humidity, and occupancy sensors) deployed throughout the M5 building. Energy data were recorded from devices like coffee machine, microwave, etc., and environmental data were captured in key areas (Interdisciplinary lab, kitchen, and mail room).

<https://github.com/adel8641/Dataset-of-IoT-Based-Energy-and-Environmental-Parameters-in-a-Smart-Building-Infrastructure>

5 (PDF) Dataset of IoT-Based Energy and Environmental Parameters in a Smart Building Infrastructure

[https://www.researchgate.net/publication/382666251\\_Dataset\\_of\\_IoT-](https://www.researchgate.net/publication/382666251_Dataset_of_IoT-Based_Energy_and_Environmental_Parameters_in_a_Smart_Building_Infrastructure)

[Based\\_Energy\\_and\\_Environmental\\_Parameters\\_in\\_a\\_Smart\\_Building\\_Infrastructure](https://www.researchgate.net/publication/382666251_Dataset_of_IoT-Based_Energy_and_Environmental_Parameters_in_a_Smart_Building_Infrastructure)

6 7 10 Dataset of IoT-based energy and environmental parameters in a smart building infrastructure - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11639320/>

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